

Supplementary Information

Harnessing Machine Learning Models to Repurpose Drugs Targeting HIV-1 Integrase, Protease, and Reverse Transcriptase

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The applicability domain, defined as the range of values for the specified molecular descriptors, such as molecular weight (MW), the partition coefficients (SlogP), topological polar surface area (TPSA), hydrogen bond donors (HBD), hydrogen bond acceptors (HBA), and molecular flexibility (RBN), was depicted using Pandas, Matplotlib, and Seaborn Python libraries for each split. The chemical spaces for the training, test, and validation sets largely overlap, as displayed in Figures [S1](#), [S2](#), [S3](#). Based on this analysis, we assume the prediction results for the test and IN descriptors validation set are considered reliable and trustworthy.

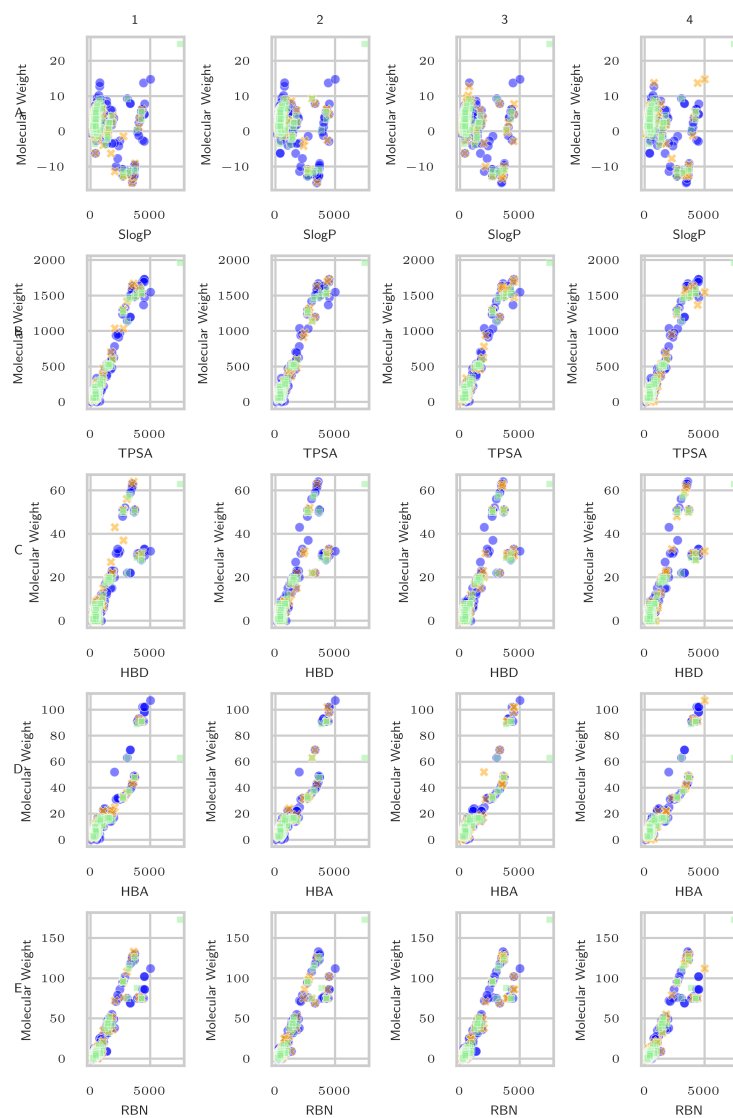


Figure S1. Chemical space analysis of the training (blue), test (orange), and validation (green) sets for HIV-1 IN datasets, using physicochemical descriptors: MW and SlogP, TPSA, HBD, HBA, and RBN.

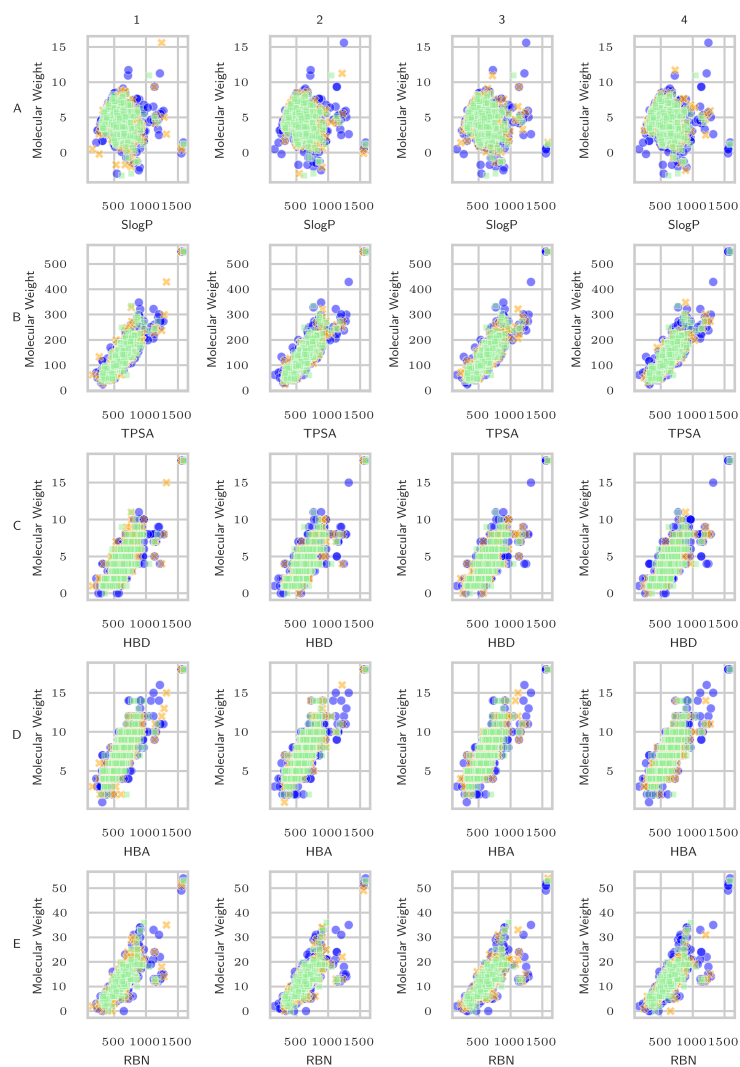


Figure S2. Chemical space analysis of the training (blue), test (orange), and validation (green) sets for HIV-1 PR datasets, using physicochemical descriptors: MW and SlogP, TPSA, HBD, HBA, and RBN.

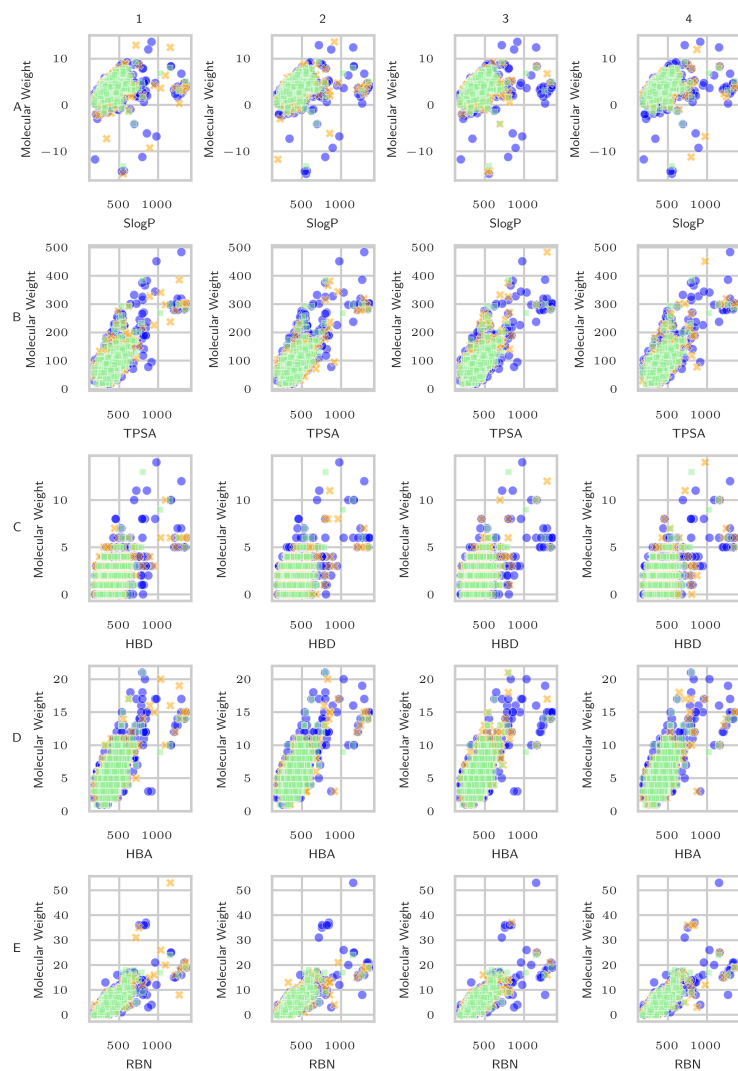


Figure S3. Chemical space analysis of the training (blue), test (orange), and validation (green) sets for HIV-1 RT datasets, using physicochemical descriptors: MW and SlogP, TPSA, HBD, HBA, and RBN.

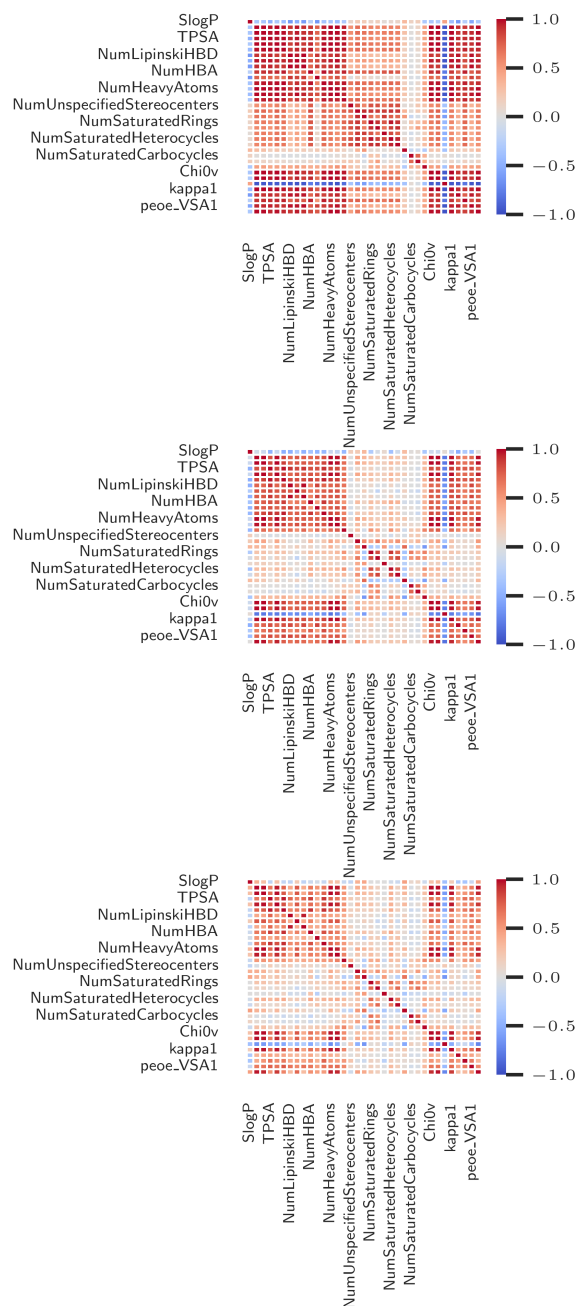


Figure S4. Heatmap plot showing the correlation matrix between descriptors; the color gradients indicate positive (red) and negative correlations (blue)

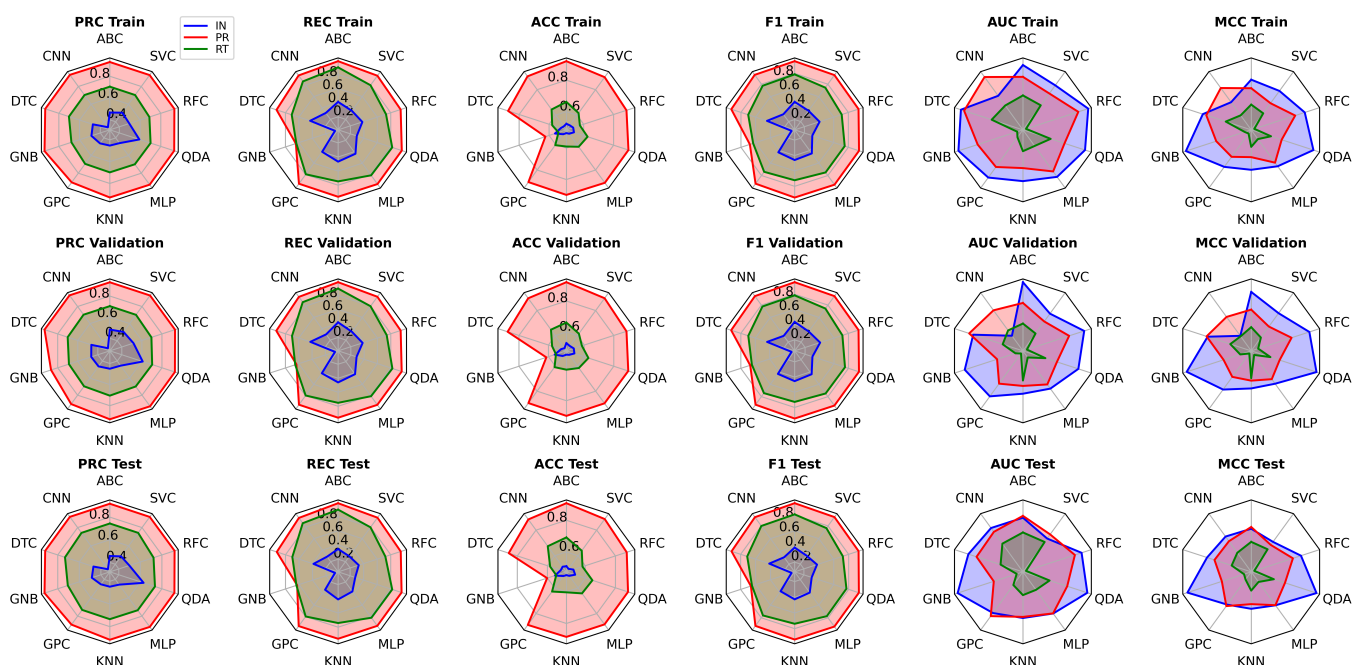


Figure S5. Performance metrics for the training, validation and test sets of the HIV-1 enzymes on PC descriptors

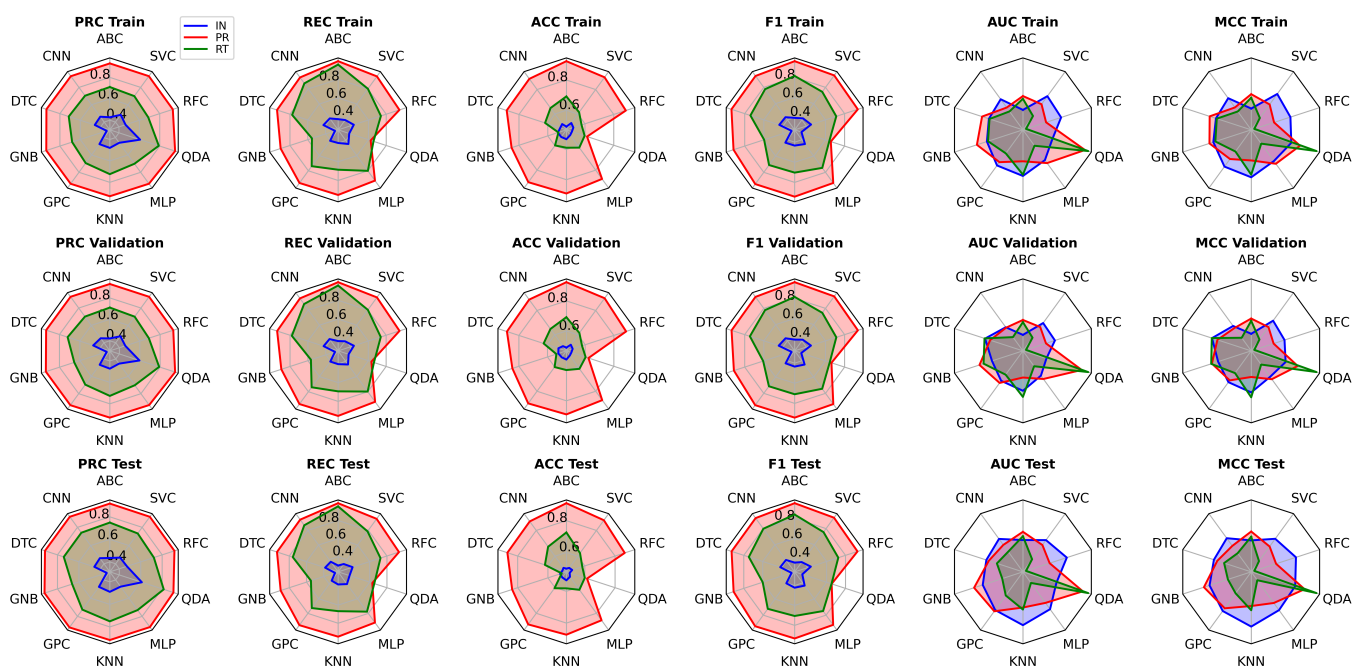


Figure S6. Performance metrics for the training, validation and test sets of the HIV-1 enzymes on ECFP4 descriptors

Table S1. Accuracy distribution for ten model (PC descriptors) - Training set

Enzyme	Model	PRC	REC	ACC	F1-Score	AUC	MCC
HIV-1 IN	GNB	0.4859	0.0621	0.5607	0.1090	0.5157	0.0694
HIV-1 IN	QDA	0.5931	0.2942	0.5353	0.3077	0.5136	0.0620
HIV-1 IN	RFC	0.4961	0.3771	0.5302	0.4069	0.5164	0.0412
HIV-1 IN	ABC	0.4672	0.4120	0.5232	0.4307	0.5132	0.0272
HIV-1 IN	DTC	0.4718	0.4321	0.5215	0.4442	0.5135	0.0287
HIV-1 IN	SVC	0.5031	0.3486	0.5205	0.3731	0.5050	0.0224
HIV-1 IN	GPC	0.4616	0.4004	0.5190	0.4274	0.5083	0.0171
HIV-1 IN	MLP	0.4592	0.4342	0.5149	0.4455	0.5076	0.0154
HIV-1 IN	CNN	0.3383	0.3118	0.5123	0.3167	0.4942	-0.0094
HIV-1 IN	KNN	0.4526	0.4661	0.5054	0.4586	0.5019	0.0038
HIV-1 PR	ABC	0.9304	0.9886	0.9206	0.9585	0.5033	0.0076
HIV-1 PR	CNN	0.9318	0.9683	0.9045	0.9496	0.5136	0.0307
HIV-1 PR	MLP	0.9302	0.9652	0.9003	0.9473	0.5020	0.0052
HIV-1 PR	QDA	0.9293	0.9649	0.8991	0.9459	0.4956	-0.0143
HIV-1 PR	SVC	0.9295	0.9635	0.8981	0.9460	0.4966	-0.0123
HIV-1 PR	KNN	0.9287	0.9640	0.8977	0.9460	0.4910	-0.0251
HIV-1 PR	GPC	0.9037	0.9593	0.8946	0.9307	0.4974	-0.0113
HIV-1 PR	RFC	0.9310	0.9476	0.8859	0.9392	0.5077	0.0182
HIV-1 PR	DTC	0.9314	0.9328	0.8737	0.9319	0.5107	0.0204
HIV-1 PR	GNB	0.9382	0.6452	0.6247	0.6781	0.4988	-0.0035
HIV-1 RT	ABC	0.7045	0.9009	0.6620	0.7870	0.4878	-0.0298
HIV-1 RT	CNN	0.7033	0.8638	0.6443	0.7744	0.4842	-0.0465
HIV-1 RT	QDA	0.7015	0.8219	0.6262	0.7504	0.4834	-0.0399
HIV-1 RT	MLP	0.6988	0.8120	0.6175	0.7503	0.4757	-0.0614
HIV-1 RT	SVC	0.7031	0.7971	0.6163	0.7442	0.4844	-0.0408
HIV-1 RT	GPC	0.6940	0.7940	0.6048	0.7405	0.4668	-0.0771
HIV-1 RT	KNN	0.6983	0.7462	0.5905	0.7213	0.4769	-0.0486
HIV-1 RT	DTC	0.7026	0.7065	0.5793	0.7010	0.4865	-0.0263
HIV-1 RT	RFC	0.6899	0.7299	0.5749	0.7093	0.4618	-0.0800
HIV-1 RT	GNB	0.6788	0.6377	0.5373	0.6334	0.4641	-0.0719

Table S2. Accuracy distribution for ten model (PC descriptors) - Validation set

Enzyme	Model	PRC	REC	ACC	F1-Score	AUC	MCC
HIV-1 IN	GNB	0.4659	0.0557	0.5578	0.0985	0.5126	0.0580
HIV-1 IN	QDA	0.6082	0.2865	0.5328	0.3015	0.5105	0.0597
HIV-1 IN	RFC	0.5164	0.3775	0.5287	0.4083	0.5150	0.0449
HIV-1 IN	ABC	0.4824	0.4164	0.5282	0.4367	0.5181	0.0405
HIV-1 IN	GPC	0.4625	0.4046	0.5193	0.4301	0.5090	0.0186
HIV-1 IN	SVC	0.5023	0.3453	0.5170	0.3701	0.5015	0.0161
HIV-1 IN	DTC	0.4649	0.4260	0.5141	0.4364	0.5061	0.0146
HIV-1 IN	MLP	0.4542	0.4207	0.5101	0.4355	0.5020	0.0044
HIV-1 IN	KNN	0.4500	0.4595	0.5031	0.4538	0.4992	-0.0016
HIV-1 IN	CNN	0.3116	0.2998	0.4995	0.2998	0.4815	-0.0391
HIV-1 PR	ABC	0.9303	0.9896	0.9214	0.9589	0.5029	0.0056
HIV-1 PR	QDA	0.9291	0.9642	0.8983	0.9454	0.4940	-0.0168
HIV-1 PR	SVC	0.9292	0.9639	0.8981	0.9460	0.4944	-0.0165
HIV-1 PR	KNN	0.9290	0.9639	0.8979	0.9461	0.4935	-0.0170
HIV-1 PR	CNN	0.9306	0.9616	0.8976	0.9456	0.5045	0.0070
HIV-1 PR	MLP	0.9297	0.9620	0.8970	0.9453	0.4982	-0.0067
HIV-1 PR	GPC	0.9038	0.9596	0.8948	0.9308	0.4975	-0.0123
HIV-1 PR	RFC	0.9304	0.9486	0.8862	0.9394	0.5034	0.0079
HIV-1 PR	DTC	0.9312	0.9370	0.8770	0.9338	0.5092	0.0168
HIV-1 PR	GNB	0.8662	0.6349	0.6143	0.6704	0.4877	-0.0184
HIV-1 RT	ABC	0.7043	0.8986	0.6610	0.7857	0.4878	-0.0290
HIV-1 RT	CNN	0.7042	0.8683	0.6474	0.7768	0.4863	-0.0403
HIV-1 RT	QDA	0.7025	0.8241	0.6280	0.7521	0.4851	-0.0359
HIV-1 RT	SVC	0.7027	0.7970	0.6157	0.7437	0.4835	-0.0433
HIV-1 RT	MLP	0.6979	0.8095	0.6154	0.7487	0.4738	-0.0658
HIV-1 RT	GPC	0.6958	0.7975	0.6083	0.7430	0.4702	-0.0693
HIV-1 RT	KNN	0.7050	0.7512	0.5998	0.7272	0.4895	-0.0218
HIV-1 RT	RFC	0.6950	0.7387	0.5838	0.7161	0.4709	-0.0613
HIV-1 RT	DTC	0.7004	0.7008	0.5751	0.6968	0.4834	-0.0324
HIV-1 RT	GNB	0.6879	0.6527	0.5491	0.6470	0.4736	-0.0530

Table S3. Accuracy distribution for ten model (PC descriptors) - Test set

Enzyme	Model	PRC	REC	ACC	F1-Score	AUC	MCC
HIV-1 IN	GNB	0.4840	0.0681	0.5497	0.1180	0.5154	0.0666
HIV-1 IN	QDA	0.6300	0.2753	0.5326	0.3047	0.5143	0.0714
HIV-1 IN	RFC	0.4982	0.3274	0.5210	0.3761	0.5072	0.0230
HIV-1 IN	CNN	0.3458	0.2863	0.5146	0.3059	0.4984	-0.0028
HIV-1 IN	DTC	0.4689	0.3861	0.5110	0.4170	0.5021	0.0055
HIV-1 IN	ABC	0.4570	0.3436	0.5102	0.3861	0.4983	-0.0044
HIV-1 IN	GPC	0.4524	0.3334	0.5069	0.3820	0.4945	-0.0120
HIV-1 IN	MLP	0.4563	0.3635	0.5053	0.4032	0.4952	-0.0096
HIV-1 IN	SVC	0.4861	0.2920	0.4977	0.3330	0.4831	-0.0223
HIV-1 IN	KNN	0.4478	0.4095	0.4950	0.4268	0.4889	-0.0227
HIV-1 PR	ABC	0.9180	0.9877	0.9077	0.9514	0.5005	0.0009
HIV-1 PR	GPC	0.8922	0.9646	0.8881	0.9270	0.4990	-0.0068
HIV-1 PR	SVC	0.9162	0.9643	0.8863	0.9395	0.4891	-0.0335
HIV-1 PR	KNN	0.9159	0.9628	0.8848	0.9387	0.4876	-0.0366
HIV-1 PR	MLP	0.9171	0.9605	0.8841	0.9381	0.4950	-0.0101
HIV-1 PR	QDA	0.9161	0.9610	0.8836	0.9371	0.4894	-0.0269
HIV-1 PR	CNN	0.9167	0.9599	0.8832	0.9377	0.4928	-0.0199
HIV-1 PR	RFC	0.9177	0.9486	0.8748	0.9329	0.4989	-0.0022
HIV-1 PR	DTC	0.9166	0.9284	0.8567	0.9221	0.4920	-0.0178
HIV-1 PR	GNB	0.9227	0.6280	0.6021	0.6627	0.4707	-0.0408
HIV-1 RT	ABC	0.7436	0.8974	0.6917	0.8092	0.4810	-0.0433
HIV-1 RT	CNN	0.7423	0.8680	0.6742	0.7993	0.4757	-0.0622
HIV-1 RT	QDA	0.7357	0.8222	0.6470	0.7704	0.4676	-0.0617
HIV-1 RT	MLP	0.7355	0.8193	0.6422	0.7743	0.4608	-0.0953
HIV-1 RT	SVC	0.7424	0.7970	0.6395	0.7652	0.4781	-0.0495
HIV-1 RT	GPC	0.7319	0.7979	0.6279	0.7633	0.4538	-0.1039
HIV-1 RT	KNN	0.7343	0.7416	0.6035	0.7378	0.4620	-0.0767
HIV-1 RT	DTC	0.7327	0.6990	0.5821	0.7120	0.4623	-0.0714
HIV-1 RT	RFC	0.7225	0.7209	0.5814	0.7217	0.4385	-0.1226
HIV-1 RT	GNB	0.7066	0.6385	0.5416	0.6478	0.4424	-0.1099

Table S4. Accuracy distribution for ten model (ECFP4 descriptors) - Training set

Enzyme	Model	PRC	REC	ACC	F1-Score	AUC	MCC
HIV-1 IN	KNN	0.4788	0.3991	0.5327	0.4343	0.5207	0.0428
HIV-1 IN	GNB	0.3408	0.2936	0.5297	0.3102	0.5084	0.0194
HIV-1 IN	GPC	0.4751	0.4035	0.5289	0.4353	0.5176	0.0367
HIV-1 IN	SVC	0.4836	0.3884	0.5265	0.4188	0.5140	0.0333
HIV-1 IN	RFC	0.4645	0.4353	0.5193	0.4494	0.5118	0.0238
HIV-1 IN	CNN	0.4618	0.4098	0.5181	0.4331	0.5083	0.0173
HIV-1 IN	MLP	0.4596	0.4458	0.5138	0.4521	0.5077	0.0156
HIV-1 IN	DTC	0.4578	0.4237	0.5134	0.4384	0.5053	0.0110
HIV-1 IN	QDA	0.6070	0.3837	0.5092	0.3491	0.4979	0.0266
HIV-1 IN	ABC	0.4277	0.3729	0.4924	0.3969	0.4816	-0.0374
HIV-1 PR	ABC	0.9302	0.9932	0.9244	0.9606	0.5023	0.0067
HIV-1 PR	SVC	0.9298	0.9655	0.9001	0.9472	0.4991	-0.0065
HIV-1 PR	GPC	0.9323	0.9613	0.8982	0.9444	0.5115	0.0079
HIV-1 PR	KNN	0.9298	0.9551	0.8912	0.9422	0.4988	-0.0089
HIV-1 PR	CNN	0.9290	0.9515	0.8872	0.9400	0.4930	-0.0178
HIV-1 PR	DTC	0.9322	0.9455	0.8853	0.9386	0.5160	0.0312
HIV-1 PR	RFC	0.9283	0.9485	0.8839	0.9383	0.4882	-0.0290
HIV-1 PR	MLP	0.9318	0.9300	0.8716	0.9308	0.5132	0.0259
HIV-1 PR	GNB	0.9338	0.9049	0.8515	0.9169	0.5241	0.0319
HIV-1 PR	QDA	0.9510	0.6251	0.6145	0.6557	0.5493	0.0548
HIV-1 RT	ABC	0.7100	0.9569	0.6917	0.8139	0.4982	-0.0042
HIV-1 RT	CNN	0.7038	0.8729	0.6491	0.7778	0.4859	-0.0329
HIV-1 RT	DTC	0.7133	0.7755	0.6187	0.7403	0.5043	0.0083
HIV-1 RT	MLP	0.7014	0.8008	0.6160	0.7451	0.4812	-0.0504
HIV-1 RT	SVC	0.6994	0.8015	0.6141	0.7450	0.4774	-0.0577
HIV-1 RT	GPC	0.6937	0.7392	0.5981	0.7105	0.4953	-0.0205
HIV-1 RT	QDA	0.7907	0.6568	0.5976	0.6400	0.5544	0.1079
HIV-1 RT	KNN	0.7237	0.6851	0.5889	0.7031	0.5188	0.0348
HIV-1 RT	RFC	0.6876	0.7401	0.5762	0.7129	0.4567	-0.0927
HIV-1 RT	GNB	0.6800	0.5529	0.5262	0.5969	0.5067	0.0129

Table S5. Accuracy distribution for ten model (ECFP4 descriptors) - Validation set

Enzyme	Model	PRC	REC	ACC	F1-Score	AUC	MCC
HIV-1 IN	KNN	0.4720	0.4033	0.5261	0.4340	0.5151	0.0315
HIV-1 IN	GNB	0.3364	0.2966	0.5244	0.3101	0.5039	0.0107
HIV-1 IN	DTC	0.4693	0.4242	0.5239	0.4444	0.5149	0.0304
HIV-1 IN	GPC	0.4666	0.4032	0.5213	0.4311	0.5107	0.0227
HIV-1 IN	SVC	0.4743	0.3877	0.5185	0.4148	0.5067	0.0183
HIV-1 IN	RFC	0.4580	0.4387	0.5122	0.4480	0.5056	0.0114
HIV-1 IN	MLP	0.4533	0.4414	0.5069	0.4460	0.5010	0.0024
HIV-1 IN	CNN	0.4511	0.4172	0.5069	0.4320	0.4988	-0.0018
HIV-1 IN	QDA	0.5967	0.3796	0.5031	0.3468	0.4920	0.0149
HIV-1 IN	ABC	0.4219	0.3662	0.4900	0.3898	0.4788	-0.0440
HIV-1 PR	ABC	0.9301	0.9934	0.9244	0.9606	0.5014	0.0030
HIV-1 PR	SVC	0.9298	0.9659	0.9005	0.9474	0.4992	-0.0061
HIV-1 PR	GPC	0.9327	0.9624	0.8997	0.9453	0.5147	0.0159
HIV-1 PR	KNN	0.9292	0.9549	0.8904	0.9418	0.4950	-0.0161
HIV-1 PR	CNN	0.9297	0.9529	0.8891	0.9410	0.4980	-0.0094
HIV-1 PR	RFC	0.9287	0.9499	0.8856	0.9391	0.4913	-0.0240
HIV-1 PR	DTC	0.9309	0.9448	0.8834	0.9375	0.5069	0.0130
HIV-1 PR	MLP	0.9310	0.9327	0.8731	0.9318	0.5073	0.0114
HIV-1 PR	GNB	0.9338	0.8978	0.8454	0.9130	0.5238	0.0307
HIV-1 PR	QDA	0.9468	0.6347	0.6226	0.6625	0.5483	0.0542
HIV-1 RT	ABC	0.7099	0.9581	0.6921	0.8143	0.4981	-0.0041
HIV-1 RT	CNN	0.7010	0.8784	0.6482	0.7784	0.4803	-0.0468
HIV-1 RT	DTC	0.7200	0.7798	0.6271	0.7457	0.5158	0.0311
HIV-1 RT	MLP	0.7025	0.7978	0.6159	0.7436	0.4834	-0.0444
HIV-1 RT	SVC	0.7004	0.8017	0.6153	0.7455	0.4793	-0.0534
HIV-1 RT	QDA	0.7901	0.6633	0.6028	0.6448	0.5587	0.1151
HIV-1 RT	GPC	0.6957	0.7415	0.6014	0.7130	0.4992	-0.0119
HIV-1 RT	KNN	0.7264	0.6933	0.5957	0.7091	0.5244	0.0466
HIV-1 RT	RFC	0.6890	0.7398	0.5777	0.7134	0.4595	-0.0866
HIV-1 RT	GNB	0.6549	0.5663	0.5380	0.6069	0.5173	0.0331

Table S6. Accuracy distribution for ten model (ECFP4 descriptors) - Test set

Enzyme	Model	PRC	REC	ACC	F1-Score	AUC	MCC
HIV-1 IN	KNN	0.4888	0.3726	0.5312	0.4218	0.5199	0.0412
HIV-1 IN	GPC	0.4760	0.3543	0.5218	0.4048	0.5099	0.0208
HIV-1 IN	GNB	0.3406	0.2699	0.5196	0.2955	0.5018	0.0059
HIV-1 IN	MLP	0.4739	0.4003	0.5165	0.4335	0.5082	0.0173
HIV-1 IN	RFC	0.4711	0.4018	0.5164	0.4335	0.5082	0.0165
HIV-1 IN	CNN	0.4617	0.3715	0.5090	0.4106	0.4993	-0.0012
HIV-1 IN	SVC	0.4679	0.3407	0.5088	0.3840	0.4969	-0.0035
HIV-1 IN	DTC	0.4563	0.3844	0.5041	0.4150	0.4956	-0.0089
HIV-1 IN	QDA	0.6104	0.3666	0.4998	0.3434	0.4903	0.0139
HIV-1 IN	ABC	0.4348	0.3041	0.4986	0.3561	0.4847	-0.0338
HIV-1 PR	ABC	0.9176	0.9906	0.9097	0.9526	0.4982	-0.0052
HIV-1 PR	GPC	0.9208	0.9599	0.8864	0.9379	0.5124	0.0099
HIV-1 PR	SVC	0.9158	0.9563	0.8792	0.9355	0.4869	-0.0343
HIV-1 PR	KNN	0.9165	0.9481	0.8730	0.9319	0.4910	-0.0252
HIV-1 PR	CNN	0.9158	0.9451	0.8699	0.9302	0.4872	-0.0318
HIV-1 PR	DTC	0.9165	0.9419	0.8679	0.9289	0.4914	-0.0203
HIV-1 PR	RFC	0.9145	0.9395	0.8638	0.9268	0.4786	-0.0510
HIV-1 PR	MLP	0.9172	0.9247	0.8542	0.9209	0.4954	-0.0113
HIV-1 PR	GNB	0.9209	0.8993	0.8364	0.9075	0.5167	0.0250
HIV-1 PR	QDA	0.9051	0.6304	0.6148	0.6535	0.5358	0.0458
HIV-1 RT	ABC	0.7490	0.9555	0.7260	0.8386	0.4910	-0.0219
HIV-1 RT	CNN	0.7411	0.8683	0.6733	0.7981	0.4735	-0.0610
HIV-1 RT	SVC	0.7330	0.7858	0.6240	0.7563	0.4581	-0.0926
HIV-1 RT	DTC	0.7412	0.7587	0.6198	0.7467	0.4774	-0.0435
HIV-1 RT	MLP	0.7309	0.7758	0.6173	0.7492	0.4550	-0.1021
HIV-1 RT	GPC	0.7260	0.7270	0.6052	0.7218	0.4804	-0.0498
HIV-1 RT	QDA	0.8145	0.6532	0.5998	0.6522	0.5451	0.0862
HIV-1 RT	RFC	0.7225	0.7251	0.5832	0.7238	0.4379	-0.1249
HIV-1 RT	KNN	0.7506	0.6670	0.5817	0.7059	0.4943	-0.0116
HIV-1 RT	GNB	0.6631	0.5487	0.5089	0.6002	0.4682	-0.0564

Table S7. Accuracy distribution for ten model (MIX descriptors) - Training set

Enzyme	Model	PRC	REC	ACC	F1-Score	AUC	MCC
HIV-1 IN	RFC	0.9955	0.9527	0.9768	0.9716	0.9747	0.9554
HIV-1 IN	KNN	0.9785	0.9213	0.9559	0.9477	0.9528	0.9128
HIV-1 IN	MLP	0.9367	0.8941	0.9240	0.9133	0.9213	0.8486
HIV-1 IN	GPC	0.9616	0.8521	0.9183	0.9030	0.9123	0.8382
HIV-1 IN	CNN	0.9164	0.8788	0.9096	0.8958	0.9068	0.8184
HIV-1 IN	SVC	0.9404	0.8310	0.8977	0.8665	0.8917	0.8007
HIV-1 IN	DTC	0.8795	0.8006	0.8616	0.8315	0.8562	0.7237
HIV-1 IN	ABC	0.7744	0.6678	0.7601	0.7117	0.7517	0.5176
HIV-1 IN	GNB	0.6891	0.4617	0.7380	0.5237	0.7131	0.4593
HIV-1 IN	QDA	0.7344	0.5012	0.6545	0.4978	0.6406	0.3131
HIV-1 PR	RFC	0.9942	1.0000	0.9946	0.9971	0.9612	0.9567
HIV-1 PR	MLP	0.9966	0.9969	0.9939	0.9967	0.9754	0.9534
HIV-1 PR	CNN	0.9893	0.9999	0.9898	0.9946	0.9281	0.9184
HIV-1 PR	KNN	0.9788	0.9982	0.9777	0.9883	0.8523	0.7915
HIV-1 PR	DTC	0.9818	0.9929	0.9760	0.9872	0.8722	0.7897
HIV-1 PR	SVC	0.9738	0.9992	0.9736	0.9861	0.8167	0.7149
HIV-1 PR	ABC	0.9472	0.9949	0.9431	0.9703	0.6255	0.3267
HIV-1 PR	GPC	0.9403	1.0000	0.9404	0.9691	0.5751	0.2493
HIV-1 PR	GNB	0.9763	0.9323	0.9135	0.9511	0.7982	0.5213
HIV-1 PR	QDA	0.8152	0.7177	0.7052	0.7188	0.6282	0.2184
HIV-1 RT	RFC	0.9936	0.9984	0.9943	0.9960	0.9912	0.9860
HIV-1 RT	KNN	0.9632	0.9720	0.9533	0.9674	0.9397	0.8854
HIV-1 RT	MLP	0.9499	0.9803	0.9458	0.9638	0.9206	0.8630
HIV-1 RT	SVC	0.8974	0.9818	0.8994	0.9353	0.8393	0.7410
HIV-1 RT	GPC	0.8656	0.9943	0.8833	0.9247	0.8023	0.7003
HIV-1 RT	DTC	0.8983	0.9071	0.8543	0.8992	0.8158	0.6287
HIV-1 RT	CNN	0.8421	0.9899	0.8521	0.9077	0.7516	0.5975
HIV-1 RT	GNB	0.9113	0.7679	0.7667	0.7881	0.7659	0.5013
HIV-1 RT	ABC	0.7590	0.9610	0.7469	0.8443	0.5908	0.2117
HIV-1 RT	QDA	0.7072	0.8090	0.7225	0.7437	0.6594	0.3111

Table S8. Accuracy distribution for ten model (MIX descriptors) - Validation set

Enzyme	Model	PRC	REC	ACC	F1-Score	AUC	MCC
HIV-1 IN	RFC	0.8680	0.6964	0.8150	0.7677	0.8043	0.6331
HIV-1 IN	KNN	0.8631	0.6359	0.7881	0.7265	0.7744	0.5827
HIV-1 IN	MLP	0.7897	0.7308	0.7846	0.7472	0.7796	0.5743
HIV-1 IN	GPC	0.8156	0.6627	0.7758	0.7253	0.7655	0.5530
HIV-1 IN	SVC	0.8130	0.5830	0.7400	0.6448	0.7259	0.4864
HIV-1 IN	CNN	0.7303	0.6737	0.7377	0.6947	0.7319	0.4736
HIV-1 IN	ABC	0.7127	0.5657	0.6955	0.6195	0.6838	0.3877
HIV-1 IN	DTC	0.6860	0.6079	0.6950	0.6376	0.6871	0.3839
HIV-1 IN	GNB	0.5587	0.3320	0.6539	0.3927	0.6248	0.2757
HIV-1 IN	QDA	0.5844	0.5404	0.5648	0.4306	0.5626	0.1591
HIV-1 PR	MLP	0.9668	0.9877	0.9568	0.9770	0.7675	0.6232
HIV-1 PR	CNN	0.9555	0.9951	0.9523	0.9749	0.6894	0.5505
HIV-1 PR	RFC	0.9484	0.9956	0.9454	0.9714	0.6377	0.4609
HIV-1 PR	KNN	0.9462	0.9959	0.9434	0.9704	0.6216	0.4175
HIV-1 PR	SVC	0.9454	0.9942	0.9410	0.9691	0.6149	0.3498
HIV-1 PR	GPC	0.9340	0.9989	0.9331	0.9653	0.5299	0.0781
HIV-1 PR	ABC	0.9403	0.9912	0.9330	0.9650	0.5760	0.1918
HIV-1 PR	DTC	0.9518	0.9686	0.9250	0.9600	0.6574	0.3445
HIV-1 PR	GNB	0.9459	0.9250	0.8807	0.9334	0.6090	0.2451
HIV-1 PR	QDA	0.8514	0.5627	0.5640	0.6026	0.5716	0.0952
HIV-1 RT	RFC	0.8278	0.9342	0.8151	0.8777	0.7282	0.5203
HIV-1 RT	MLP	0.8170	0.8688	0.7644	0.8393	0.6883	0.4081
HIV-1 RT	GPC	0.7648	0.9664	0.7628	0.8526	0.6143	0.3466
HIV-1 RT	SVC	0.7846	0.9243	0.7599	0.8456	0.6401	0.3451
HIV-1 RT	KNN	0.8478	0.7976	0.7515	0.8178	0.7179	0.4305
HIV-1 RT	CNN	0.7513	0.9334	0.7303	0.8310	0.5823	0.2267
HIV-1 RT	ABC	0.7457	0.9250	0.7150	0.8192	0.5619	0.1408
HIV-1 RT	DTC	0.7905	0.7658	0.6879	0.7710	0.6310	0.2610
HIV-1 RT	GNB	0.7799	0.6515	0.6464	0.6876	0.6427	0.2644
HIV-1 RT	QDA	0.7143	0.5569	0.5571	0.5456	0.5572	0.1257

Table S9. Accuracy distribution for ten model (MIX descriptors) - Test set

Enzyme	Model	PRC	REC	ACC	F1-Score	AUC	MCC
HIV-1 IN	RFC	0.9545	0.7342	0.8606	0.8283	0.8516	0.7334
HIV-1 IN	MLP	0.8741	0.8007	0.8497	0.8283	0.8462	0.7063
HIV-1 IN	GPC	0.9247	0.7367	0.8491	0.8177	0.8411	0.7073
HIV-1 IN	KNN	0.9509	0.7035	0.8452	0.8052	0.8351	0.7064
HIV-1 IN	CNN	0.8061	0.7592	0.7997	0.7751	0.7969	0.6038
HIV-1 IN	SVC	0.8971	0.6503	0.7978	0.7171	0.7873	0.6105
HIV-1 IN	ABC	0.8088	0.6518	0.7643	0.7170	0.7563	0.5326
HIV-1 IN	DTC	0.7610	0.7003	0.7579	0.7247	0.7538	0.5155
HIV-1 IN	GNB	0.5913	0.3403	0.6807	0.4205	0.6565	0.3495
HIV-1 IN	QDA	0.5345	0.5625	0.5853	0.4540	0.5837	0.2017
HIV-1 PR	RFC	0.9607	0.9864	0.9504	0.9733	0.7671	0.6290
HIV-1 PR	MLP	0.9628	0.9776	0.9444	0.9700	0.7756	0.6016
HIV-1 PR	CNN	0.9464	0.9901	0.9393	0.9677	0.6807	0.5013
HIV-1 PR	KNN	0.9455	0.9855	0.9337	0.9647	0.6697	0.4015
HIV-1 PR	SVC	0.9395	0.9906	0.9321	0.9641	0.6347	0.3434
HIV-1 PR	GPC	0.9232	0.9981	0.9217	0.9591	0.5332	0.0732
HIV-1 PR	ABC	0.9371	0.9745	0.9152	0.9545	0.6136	0.2296
HIV-1 PR	DTC	0.9641	0.9433	0.9152	0.9531	0.7717	0.4818
HIV-1 PR	GNB	0.9292	0.9369	0.8761	0.9318	0.5663	0.1219
HIV-1 PR	QDA	0.7991	0.5268	0.5314	0.5590	0.5550	0.0664
HIV-1 RT	RFC	0.8581	0.9417	0.8388	0.8979	0.7335	0.5323
HIV-1 RT	SVC	0.8288	0.9341	0.7993	0.8753	0.6612	0.3817
HIV-1 RT	GPC	0.8044	0.9721	0.7989	0.8792	0.6215	0.3689
HIV-1 RT	MLP	0.8603	0.8770	0.7962	0.8660	0.7135	0.4396
HIV-1 RT	KNN	0.8917	0.8179	0.7865	0.8513	0.7543	0.4802
HIV-1 RT	CNN	0.7914	0.9463	0.7700	0.8612	0.5894	0.2419
HIV-1 RT	ABC	0.7840	0.9485	0.7591	0.8545	0.5650	0.1531
HIV-1 RT	DTC	0.8363	0.7810	0.7170	0.8030	0.6514	0.2857
HIV-1 RT	GNB	0.7952	0.6456	0.6572	0.7045	0.6691	0.2943
HIV-1 RT	QDA	0.7602	0.5517	0.5537	0.5518	0.5558	0.1210

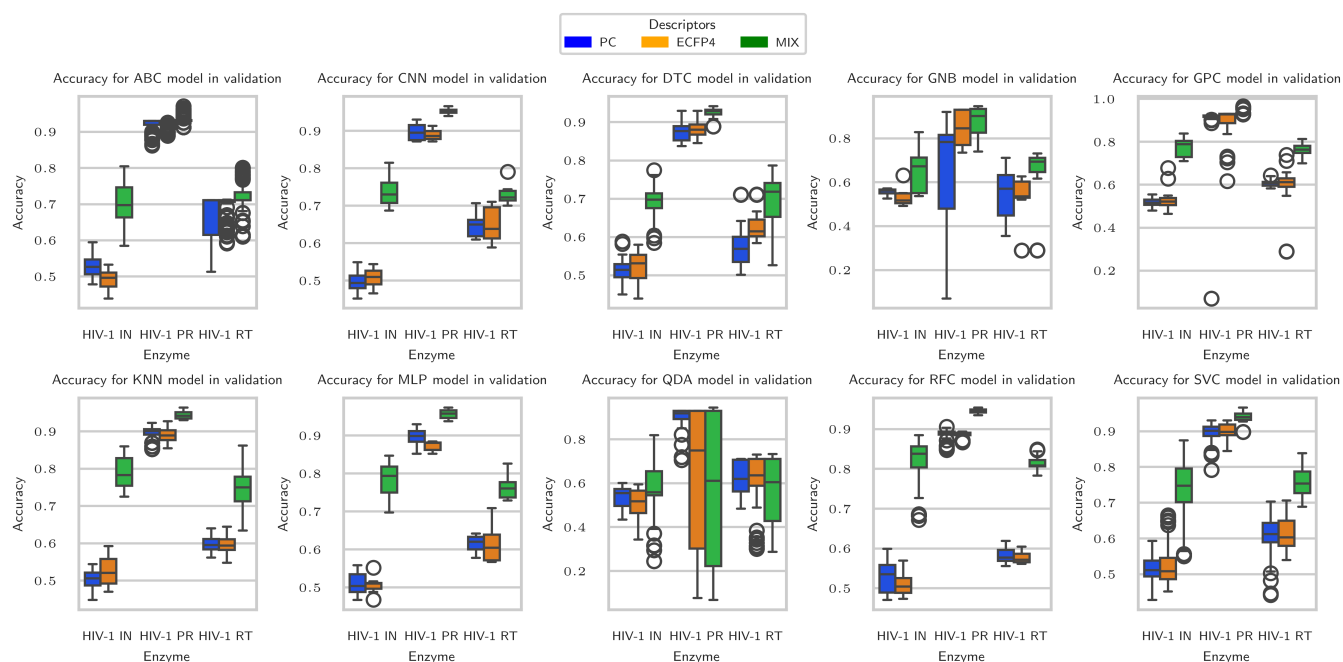


Figure S7. Accuracy distribution on the ChEMBL dataset (validation set), based on PC, ECFP4, and MIX descriptors on HIV-1 enzyme targets IN, PR, RT for ABC, CNN, DTC, GNB, GPC, KNN, MLP, QDA, RFC, and SVC models

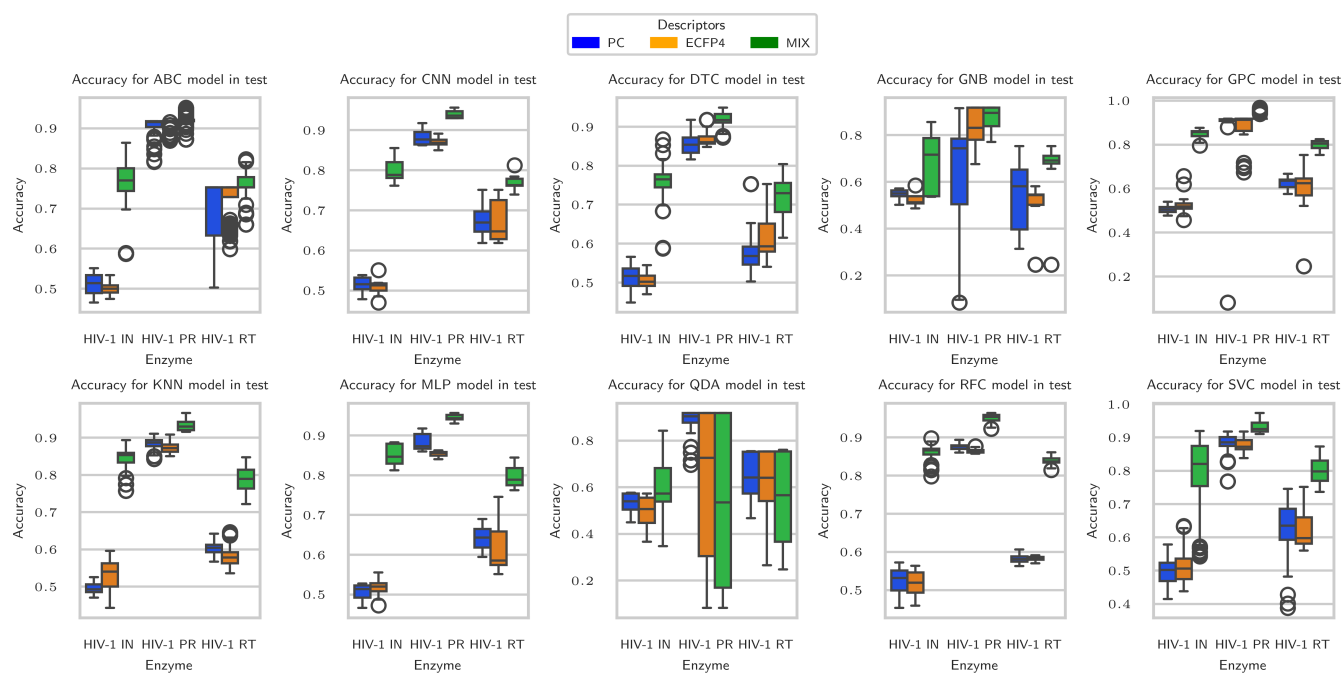


Figure S8. Accuracy distribution on the ChEMBL dataset (test set), based on PC, ECFP4, and MIX descriptors on HIV-1 enzyme targets IN, PR, RT for ABC, CNN, DTC, GNB, GPC, KNN, MLP, QDA, RFC, and SVC models

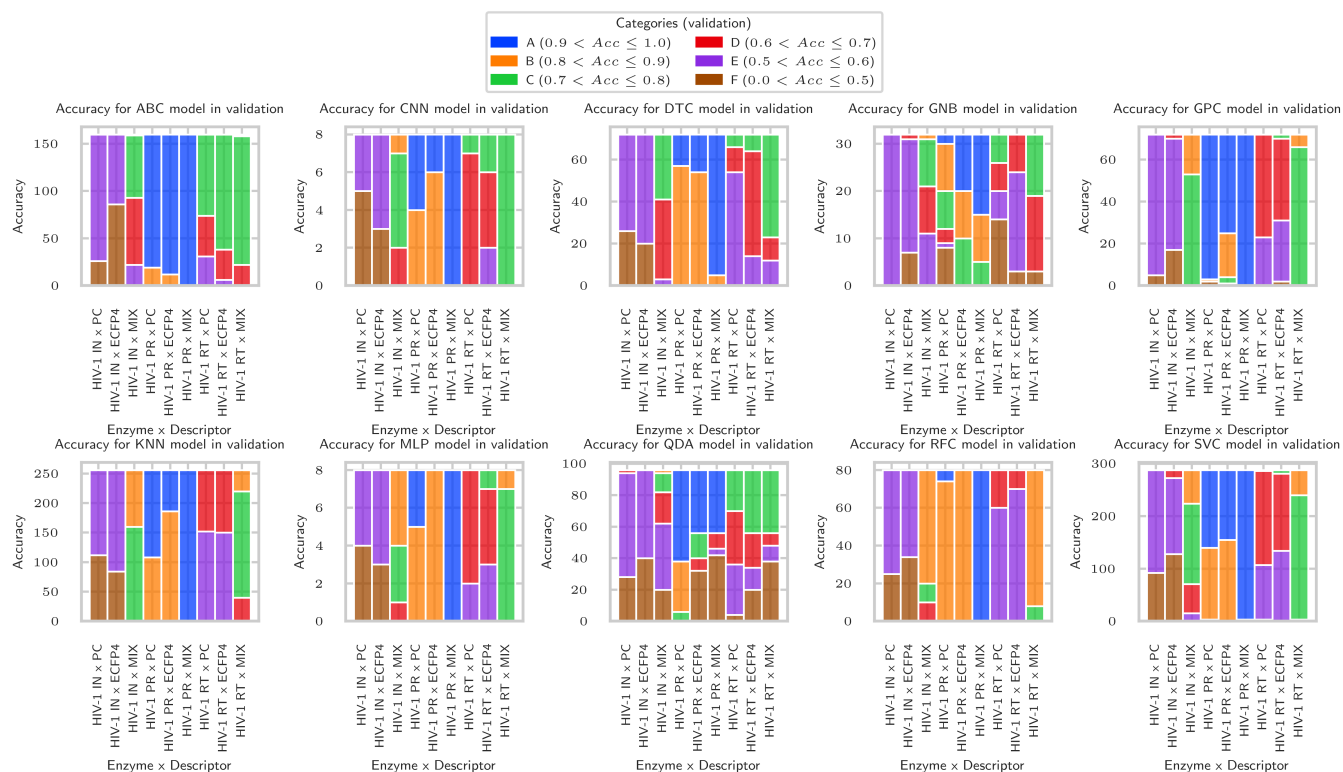


Figure S9. Accuracy categories using the prediction ChEMBL validation set displayed in Cartesian product order: enzyme targets IN, PR, and RT x descriptors PC, ECFP4, and MIX

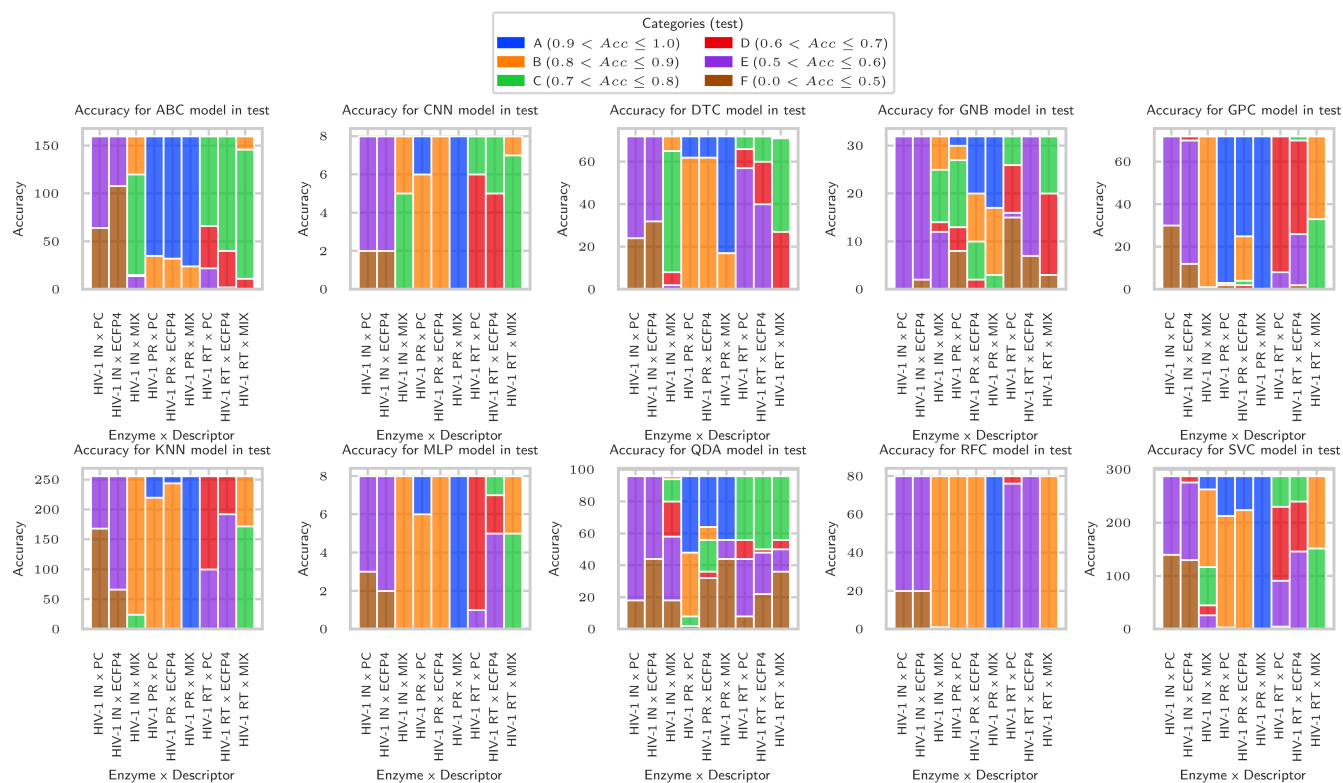


Figure S10. Accuracy categories using the prediction ChEMBL test set displayed in Cartesian product order: enzyme targets IN, PR, and RT x descriptors PC, ECFP4, and MIX

Table S10. Accuracy distribution for ABC model - Training set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5232	0.4805	0.5706	0.0232
HIV-1 IN	ECFP4	0.4924	0.4581	0.5287	0.0209
HIV-1 IN	MIX	0.7601	0.6409	0.8964	0.0595
HIV-1 PR	PC	0.9206	0.8558	0.9300	0.0147
HIV-1 PR	ECFP4	0.9244	0.8936	0.9300	0.0107
HIV-1 PR	MIX	0.9431	0.9299	0.9976	0.0204
HIV-1 RT	PC	0.6620	0.5320	0.7109	0.0608
HIV-1 RT	ECFP4	0.6917	0.6076	0.7114	0.0339
HIV-1 RT	MIX	0.7469	0.6983	0.8702	0.0501

Table S11. Accuracy distribution for ABC model - Validation set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5282	0.4787	0.5949	0.0280
HIV-1 IN	ECFP4	0.4900	0.4392	0.5330	0.0260
HIV-1 IN	MIX	0.6955	0.5851	0.8043	0.0576
HIV-1 PR	PC	0.9214	0.8623	0.9300	0.0150
HIV-1 PR	ECFP4	0.9244	0.8889	0.9300	0.0112
HIV-1 PR	MIX	0.9330	0.9128	0.9710	0.0117
HIV-1 RT	PC	0.6610	0.5135	0.7111	0.0637
HIV-1 RT	ECFP4	0.6921	0.5913	0.7121	0.0359
HIV-1 RT	MIX	0.7150	0.6108	0.8009	0.0387

Table S12. Accuracy distribution for ABC model - Test set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5102	0.4660	0.5511	0.0245
HIV-1 IN	ECFP4	0.4986	0.4745	0.5340	0.0122
HIV-1 IN	MIX	0.7643	0.5872	0.8638	0.0660
HIV-1 PR	PC	0.9077	0.8164	0.9179	0.0183
HIV-1 PR	ECFP4	0.9097	0.8696	0.9179	0.0155
HIV-1 PR	MIX	0.9152	0.8720	0.9517	0.0198
HIV-1 RT	PC	0.6917	0.5030	0.7530	0.0794
HIV-1 RT	ECFP4	0.7260	0.5988	0.7530	0.0475
HIV-1 RT	MIX	0.7591	0.6602	0.8234	0.0307

Table S13. Accuracy distribution for CNN model - Training set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5123	0.4581	0.5607	0.0372
HIV-1 IN	ECFP4	0.5181	0.4847	0.5635	0.0228
HIV-1 IN	MIX	0.9096	0.8224	0.9822	0.0498
HIV-1 PR	PC	0.9045	0.8928	0.9299	0.0157
HIV-1 PR	ECFP4	0.8872	0.8646	0.9001	0.0120
HIV-1 PR	MIX	0.9898	0.9775	0.9984	0.0069
HIV-1 RT	PC	0.6443	0.6111	0.7058	0.0319
HIV-1 RT	ECFP4	0.6491	0.6036	0.7094	0.0442
HIV-1 RT	MIX	0.8521	0.7249	0.9625	0.0930

Table S14. Accuracy distribution for CNN model - Validation set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.4995	0.4520	0.5489	0.0341
HIV-1 IN	ECFP4	0.5069	0.4660	0.5437	0.0261
HIV-1 IN	MIX	0.7377	0.6872	0.8149	0.0428
HIV-1 PR	PC	0.8976	0.8717	0.9300	0.0243
HIV-1 PR	ECFP4	0.8891	0.8717	0.9130	0.0141
HIV-1 PR	MIX	0.9523	0.9396	0.9662	0.0082
HIV-1 RT	PC	0.6474	0.6093	0.7066	0.0324
HIV-1 RT	ECFP4	0.6482	0.5883	0.7106	0.0491
HIV-1 RT	MIX	0.7303	0.7001	0.7904	0.0280

Table S15. Accuracy distribution for CNN model - Test set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5146	0.4787	0.5383	0.0216
HIV-1 IN	ECFP4	0.5090	0.4702	0.5511	0.0241
HIV-1 IN	MIX	0.7997	0.7617	0.8553	0.0305
HIV-1 PR	PC	0.8832	0.8623	0.9179	0.0226
HIV-1 PR	ECFP4	0.8699	0.8502	0.8913	0.0129
HIV-1 PR	MIX	0.9393	0.9300	0.9565	0.0104
HIV-1 RT	PC	0.6742	0.6183	0.7515	0.0448
HIV-1 RT	ECFP4	0.6733	0.6183	0.7515	0.0565
HIV-1 RT	MIX	0.7700	0.7395	0.8129	0.0218

Table S16. Accuracy distribution for DTC model - Training set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5215	0.4794	0.5770	0.0210
HIV-1 IN	ECFP4	0.5134	0.4581	0.5798	0.0237
HIV-1 IN	MIX	0.8616	0.6409	0.9865	0.0974
HIV-1 PR	PC	0.8737	0.8333	0.9299	0.0264
HIV-1 PR	ECFP4	0.8853	0.8662	0.9300	0.0207
HIV-1 PR	MIX	0.9760	0.9202	0.9976	0.0225
HIV-1 RT	PC	0.5793	0.5102	0.7109	0.0509
HIV-1 RT	ECFP4	0.6187	0.5741	0.7109	0.0409
HIV-1 RT	MIX	0.8543	0.6983	0.9715	0.0826

Table S17. Accuracy distribution for DTC model - Validation set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5141	0.4499	0.5885	0.0257
HIV-1 IN	ECFP4	0.5239	0.4392	0.5800	0.0381
HIV-1 IN	MIX	0.6950	0.5851	0.7745	0.0405
HIV-1 PR	PC	0.8770	0.8378	0.9300	0.0264
HIV-1 PR	ECFP4	0.8834	0.8450	0.9300	0.0250
HIV-1 PR	MIX	0.9250	0.8889	0.9420	0.0123
HIV-1 RT	PC	0.5751	0.5015	0.7111	0.0555
HIV-1 RT	ECFP4	0.6271	0.5838	0.7111	0.0377
HIV-1 RT	MIX	0.6879	0.5262	0.7871	0.0770

Table S18. Accuracy distribution for DTC model - Test set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5110	0.4489	0.5660	0.0302
HIV-1 IN	ECFP4	0.5041	0.4702	0.5447	0.0185
HIV-1 IN	MIX	0.7579	0.5872	0.8681	0.0456
HIV-1 PR	PC	0.8567	0.8164	0.9179	0.0291
HIV-1 PR	ECFP4	0.8679	0.8478	0.9179	0.0214
HIV-1 PR	MIX	0.9152	0.8720	0.9493	0.0198
HIV-1 RT	PC	0.5821	0.5030	0.7530	0.0606
HIV-1 RT	ECFP4	0.6198	0.5404	0.7530	0.0638
HIV-1 RT	MIX	0.7170	0.6153	0.8039	0.0510

Table S19. Accuracy distribution for GNB model - Training set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5607	0.5362	0.5774	0.0098
HIV-1 IN	ECFP4	0.5297	0.5057	0.5997	0.0216
HIV-1 IN	MIX	0.7380	0.5493	0.8808	0.1396
HIV-1 PR	PC	0.6247	0.0725	0.9243	0.3168
HIV-1 PR	ECFP4	0.8515	0.7228	0.9300	0.0709
HIV-1 PR	MIX	0.9135	0.8082	0.9815	0.0541
HIV-1 RT	PC	0.5373	0.3482	0.7109	0.1225
HIV-1 RT	ECFP4	0.5262	0.2891	0.5726	0.0780
HIV-1 RT	MIX	0.7667	0.2896	0.8587	0.1607

Table S20. Accuracy distribution for GNB model - Validation set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5578	0.5255	0.5714	0.0115
HIV-1 IN	ECFP4	0.5244	0.4925	0.6311	0.0280
HIV-1 IN	MIX	0.6539	0.5373	0.8277	0.0885
HIV-1 PR	PC	0.6143	0.0702	0.9203	0.3143
HIV-1 PR	ECFP4	0.8454	0.7343	0.9300	0.0751
HIV-1 PR	MIX	0.8807	0.7391	0.9469	0.0680
HIV-1 RT	PC	0.5491	0.3548	0.7111	0.1162
HIV-1 RT	ECFP4	0.5380	0.2894	0.6267	0.0900
HIV-1 RT	MIX	0.6464	0.2894	0.7305	0.1213

Table S21. Accuracy distribution for GNB model - Test set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5497	0.5021	0.5723	0.0163
HIV-1 IN	ECFP4	0.5196	0.4872	0.5851	0.0187
HIV-1 IN	MIX	0.6807	0.5362	0.8553	0.1203
HIV-1 PR	PC	0.6021	0.0845	0.9155	0.2964
HIV-1 PR	ECFP4	0.8364	0.6763	0.9179	0.0746
HIV-1 PR	MIX	0.8761	0.7705	0.9179	0.0476
HIV-1 RT	PC	0.5416	0.3144	0.7530	0.1515
HIV-1 RT	ECFP4	0.5089	0.2470	0.5808	0.0892
HIV-1 RT	MIX	0.6572	0.2470	0.7530	0.1363

Table S22. Accuracy distribution for GPC model - Training set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5190	0.4844	0.5479	0.0167
HIV-1 IN	ECFP4	0.5289	0.4925	0.6757	0.0285
HIV-1 IN	MIX	0.9183	0.8842	1.0000	0.0297
HIV-1 PR	PC	0.8946	0.0700	0.9300	0.1405
HIV-1 PR	ECFP4	0.8982	0.6624	0.9300	0.0598
HIV-1 PR	MIX	0.9404	0.9299	1.0000	0.0213
HIV-1 RT	PC	0.6048	0.5759	0.6409	0.0166
HIV-1 RT	ECFP4	0.5981	0.2891	0.7108	0.0595
HIV-1 RT	MIX	0.8833	0.7108	0.9990	0.0489

Table S23. Accuracy distribution for GPC model - Validation set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5193	0.4797	0.5544	0.0167
HIV-1 IN	ECFP4	0.5213	0.4648	0.6780	0.0308
HIV-1 IN	MIX	0.7758	0.7100	0.8383	0.0419
HIV-1 PR	PC	0.8948	0.0700	0.9300	0.1406
HIV-1 PR	ECFP4	0.8997	0.6174	0.9300	0.0580
HIV-1 PR	MIX	0.9331	0.9275	0.9661	0.0092
HIV-1 RT	PC	0.6083	0.5823	0.6432	0.0152
HIV-1 RT	ECFP4	0.6014	0.2889	0.7391	0.0629
HIV-1 RT	MIX	0.7628	0.7001	0.8129	0.0260

Table S24. Accuracy distribution for GPC model - Test set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5069	0.4766	0.5404	0.0177
HIV-1 IN	ECFP4	0.5218	0.4574	0.6574	0.0291
HIV-1 IN	MIX	0.8491	0.7957	0.8766	0.0194
HIV-1 PR	PC	0.8881	0.0821	0.9179	0.1373
HIV-1 PR	ECFP4	0.8864	0.6739	0.9179	0.0537
HIV-1 PR	MIX	0.9217	0.9179	0.9686	0.0114
HIV-1 RT	PC	0.6279	0.5749	0.6677	0.0201
HIV-1 RT	ECFP4	0.6052	0.2470	0.7530	0.0738
HIV-1 RT	MIX	0.7989	0.7530	0.8249	0.0192

Table S25. Accuracy distribution for KNN model - Training set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5054	0.4705	0.5305	0.0129
HIV-1 IN	ECFP4	0.5327	0.4989	0.5990	0.0271
HIV-1 IN	MIX	0.9559	0.8558	1.0000	0.0454
HIV-1 PR	PC	0.8977	0.8688	0.9194	0.0125
HIV-1 PR	ECFP4	0.8912	0.8662	0.9251	0.0164
HIV-1 PR	MIX	0.9777	0.9380	1.0000	0.0227
HIV-1 RT	PC	0.5905	0.5537	0.6344	0.0193
HIV-1 RT	ECFP4	0.5889	0.5584	0.6311	0.0164
HIV-1 RT	MIX	0.9533	0.8802	0.9995	0.0449

Table S26. Accuracy distribution for KNN model - Validation set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5031	0.4478	0.5437	0.0236
HIV-1 IN	ECFP4	0.5261	0.4702	0.5928	0.0363
HIV-1 IN	MIX	0.7881	0.7249	0.8596	0.0392
HIV-1 PR	PC	0.8979	0.8523	0.9227	0.0152
HIV-1 PR	ECFP4	0.8904	0.8547	0.9275	0.0193
HIV-1 PR	MIX	0.9434	0.9300	0.9638	0.0096
HIV-1 RT	PC	0.5998	0.5614	0.6402	0.0199
HIV-1 RT	ECFP4	0.5957	0.5479	0.6447	0.0217
HIV-1 RT	MIX	0.7515	0.6342	0.8623	0.0525

Table S27. Accuracy distribution for KNN model - Test set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.4950	0.4702	0.5255	0.0140
HIV-1 IN	ECFP4	0.5312	0.4426	0.5957	0.0410
HIV-1 IN	MIX	0.8452	0.7574	0.8936	0.0270
HIV-1 PR	PC	0.8848	0.8430	0.9106	0.0142
HIV-1 PR	ECFP4	0.8730	0.8502	0.9082	0.0137
HIV-1 PR	MIX	0.9337	0.9155	0.9662	0.0152
HIV-1 RT	PC	0.6035	0.5674	0.6422	0.0168
HIV-1 RT	ECFP4	0.5817	0.5359	0.6467	0.0284
HIV-1 RT	MIX	0.7865	0.7216	0.8473	0.0321

Table S28. Accuracy distribution for MLP model - Training set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5149	0.4918	0.5344	0.0155
HIV-1 IN	ECFP4	0.5138	0.4947	0.5415	0.0189
HIV-1 IN	MIX	0.9240	0.8630	0.9680	0.0349
HIV-1 PR	PC	0.9003	0.8784	0.9299	0.0211
HIV-1 PR	ECFP4	0.8716	0.8413	0.8873	0.0138
HIV-1 PR	MIX	0.9939	0.9879	0.9992	0.0049
HIV-1 RT	PC	0.6175	0.5949	0.6533	0.0230
HIV-1 RT	ECFP4	0.6160	0.5764	0.7034	0.0440
HIV-1 RT	MIX	0.9458	0.8013	0.9850	0.0597

Table S29. Accuracy distribution for MLP model - Validation set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5101	0.4670	0.5586	0.0332
HIV-1 IN	ECFP4	0.5069	0.4681	0.5522	0.0241
HIV-1 IN	MIX	0.7846	0.6972	0.8468	0.0519
HIV-1 PR	PC	0.8970	0.8523	0.9300	0.0264
HIV-1 PR	ECFP4	0.8731	0.8523	0.8841	0.0122
HIV-1 PR	MIX	0.9568	0.9372	0.9734	0.0133
HIV-1 RT	PC	0.6154	0.5778	0.6417	0.0240
HIV-1 RT	ECFP4	0.6159	0.5674	0.7091	0.0515
HIV-1 RT	MIX	0.7644	0.7290	0.8263	0.0335

Table S30. Accuracy distribution for MLP model - Test set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5053	0.4660	0.5277	0.0239
HIV-1 IN	ECFP4	0.5165	0.4723	0.5553	0.0264
HIV-1 IN	MIX	0.8497	0.8128	0.8830	0.0287
HIV-1 PR	PC	0.8841	0.8599	0.9179	0.0239
HIV-1 PR	ECFP4	0.8542	0.8406	0.8623	0.0075
HIV-1 PR	MIX	0.9444	0.9300	0.9565	0.0087
HIV-1 RT	PC	0.6422	0.5943	0.6901	0.0321
HIV-1 RT	ECFP4	0.6173	0.5509	0.7455	0.0705
HIV-1 RT	MIX	0.7962	0.7620	0.8443	0.0303

Table S31. Accuracy distribution for QDA model - Training set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5353	0.4517	0.5774	0.0417
HIV-1 IN	ECFP4	0.5092	0.3612	0.6111	0.0622
HIV-1 IN	MIX	0.6545	0.0021	1.0000	0.3258
HIV-1 PR	PC	0.8991	0.7309	0.9300	0.0476
HIV-1 PR	ECFP4	0.6145	0.0757	0.9300	0.3438
HIV-1 PR	MIX	0.7052	0.0451	0.9992	0.3814
HIV-1 RT	PC	0.6262	0.4890	0.7109	0.0726
HIV-1 RT	ECFP4	0.5976	0.3035	0.7109	0.1341
HIV-1 RT	MIX	0.7225	0.1618	0.9945	0.2535

Table S32. Accuracy distribution for QDA model - Validation set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5328	0.4340	0.6013	0.0478
HIV-1 IN	ECFP4	0.5031	0.3433	0.5949	0.0701
HIV-1 IN	MIX	0.5648	0.2447	0.8191	0.1268
HIV-1 PR	PC	0.8983	0.7053	0.9300	0.0504
HIV-1 PR	ECFP4	0.6226	0.0773	0.9300	0.3448
HIV-1 PR	MIX	0.5640	0.0676	0.9444	0.3537
HIV-1 RT	PC	0.6280	0.4835	0.7111	0.0708
HIV-1 RT	ECFP4	0.6028	0.3009	0.7301	0.1382
HIV-1 RT	MIX	0.5571	0.2879	0.7331	0.1643

Table S33. Accuracy distribution for QDA model - Test set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5326	0.4489	0.5766	0.0393
HIV-1 IN	ECFP4	0.4998	0.3660	0.5723	0.0620
HIV-1 IN	MIX	0.5853	0.3468	0.8426	0.1206
HIV-1 PR	PC	0.8836	0.6957	0.9179	0.0525
HIV-1 PR	ECFP4	0.6148	0.0821	0.9179	0.3358
HIV-1 PR	MIX	0.5314	0.0821	0.9179	0.3554
HIV-1 RT	PC	0.6470	0.4671	0.7545	0.0963
HIV-1 RT	ECFP4	0.5998	0.2650	0.7530	0.1649
HIV-1 RT	MIX	0.5537	0.2470	0.7605	0.1989

Table S34. Accuracy distribution for RFC model - Training set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5302	0.4979	0.5777	0.0199
HIV-1 IN	ECFP4	0.5193	0.5046	0.5557	0.0158
HIV-1 IN	MIX	0.9768	0.8417	1.0000	0.0391
HIV-1 PR	PC	0.8859	0.8711	0.9106	0.0097
HIV-1 PR	ECFP4	0.8839	0.8759	0.8905	0.0039
HIV-1 PR	MIX	0.9946	0.9807	1.0000	0.0061
HIV-1 RT	PC	0.5749	0.5457	0.6059	0.0164
HIV-1 RT	ECFP4	0.5762	0.5597	0.5939	0.0112
HIV-1 RT	MIX	0.9943	0.9750	0.9985	0.0047

Table S35. Accuracy distribution for RFC model - Validation set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5287	0.4702	0.5991	0.0408
HIV-1 IN	ECFP4	0.5122	0.4733	0.5693	0.0331
HIV-1 IN	MIX	0.8150	0.6716	0.8851	0.0624
HIV-1 PR	PC	0.8862	0.8475	0.9058	0.0139
HIV-1 PR	ECFP4	0.8856	0.8668	0.8937	0.0094
HIV-1 PR	MIX	0.9454	0.9348	0.9541	0.0049
HIV-1 RT	PC	0.5838	0.5554	0.6192	0.0186
HIV-1 RT	ECFP4	0.5777	0.5614	0.6042	0.0150
HIV-1 RT	MIX	0.8151	0.7829	0.8488	0.0166

Table S36. Accuracy distribution for RFC model - Test set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5210	0.4532	0.5723	0.0379
HIV-1 IN	ECFP4	0.5164	0.4596	0.5638	0.0365
HIV-1 IN	MIX	0.8606	0.7979	0.8979	0.0189
HIV-1 PR	PC	0.8748	0.8599	0.8937	0.0074
HIV-1 PR	ECFP4	0.8638	0.8575	0.8768	0.0061
HIV-1 PR	MIX	0.9504	0.9227	0.9638	0.0114
HIV-1 RT	PC	0.5814	0.5629	0.6063	0.0099
HIV-1 RT	ECFP4	0.5832	0.5704	0.5913	0.0050
HIV-1 RT	MIX	0.8388	0.8144	0.8608	0.0100

Table S37. Accuracy distribution for SVC model - Training set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5205	0.4318	0.5774	0.0305
HIV-1 IN	ECFP4	0.5265	0.4691	0.6579	0.0370
HIV-1 IN	MIX	0.8977	0.5881	1.0000	0.0954
HIV-1 PR	PC	0.8981	0.7907	0.9300	0.0211
HIV-1 PR	ECFP4	0.9001	0.8695	0.9300	0.0180
HIV-1 PR	MIX	0.9736	0.9299	1.0000	0.0242
HIV-1 RT	PC	0.6163	0.4540	0.6988	0.0409
HIV-1 RT	ECFP4	0.6141	0.5447	0.7023	0.0414
HIV-1 RT	MIX	0.8994	0.7374	0.9990	0.0860

Table S38. Accuracy distribution for SVC model - Validation set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.5170	0.4277	0.5928	0.0358
HIV-1 IN	ECFP4	0.5185	0.4520	0.6652	0.0441
HIV-1 IN	MIX	0.7400	0.5501	0.8745	0.0743
HIV-1 PR	PC	0.8981	0.7918	0.9300	0.0233
HIV-1 PR	ECFP4	0.9005	0.8450	0.9300	0.0205
HIV-1 PR	MIX	0.9410	0.8986	0.9661	0.0113
HIV-1 RT	PC	0.6157	0.4416	0.7031	0.0430
HIV-1 RT	ECFP4	0.6153	0.5397	0.7061	0.0434
HIV-1 RT	MIX	0.7599	0.6886	0.8383	0.0357

Table S39. Accuracy distribution for SVC model - Test set

Enzyme	Descriptor	Mean	Min	Max	Std Dev
HIV-1 IN	PC	0.4977	0.4149	0.5787	0.0433
HIV-1 IN	ECFP4	0.5088	0.4383	0.6340	0.0416
HIV-1 IN	MIX	0.7978	0.5426	0.9191	0.0989
HIV-1 PR	PC	0.8863	0.7681	0.9179	0.0222
HIV-1 PR	ECFP4	0.8792	0.8382	0.9179	0.0214
HIV-1 PR	MIX	0.9321	0.9106	0.9734	0.0148
HIV-1 RT	PC	0.6395	0.3877	0.7455	0.0640
HIV-1 RT	ECFP4	0.6240	0.5599	0.7515	0.0562
HIV-1 RT	MIX	0.7993	0.7365	0.8728	0.0335